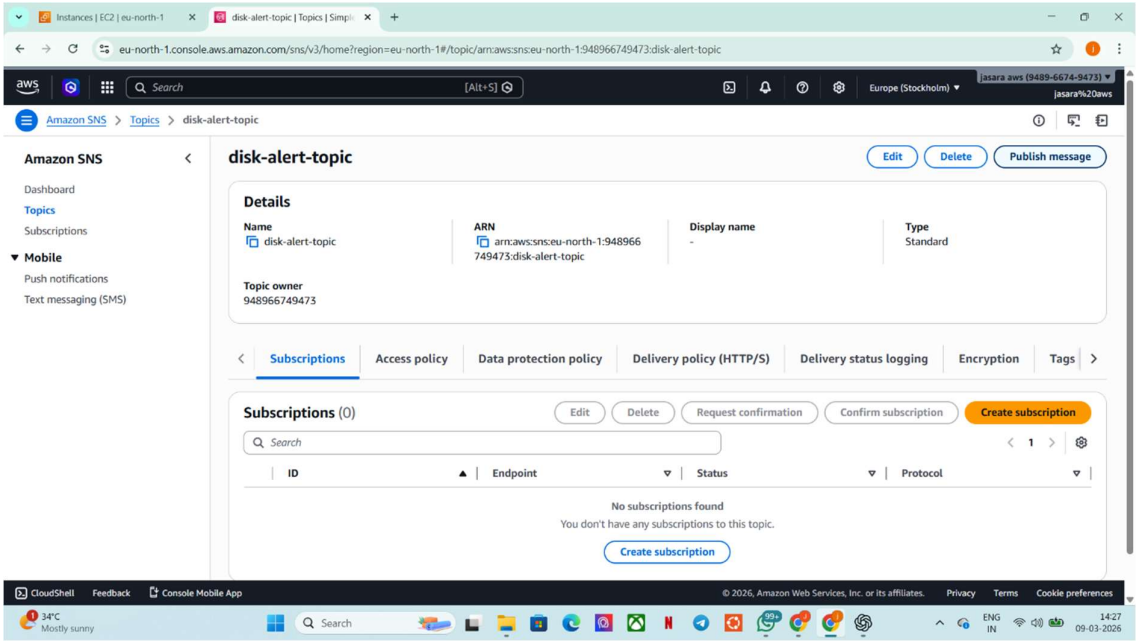
DISK SPACE UTILIZATION NOTIFICATION USING SNS

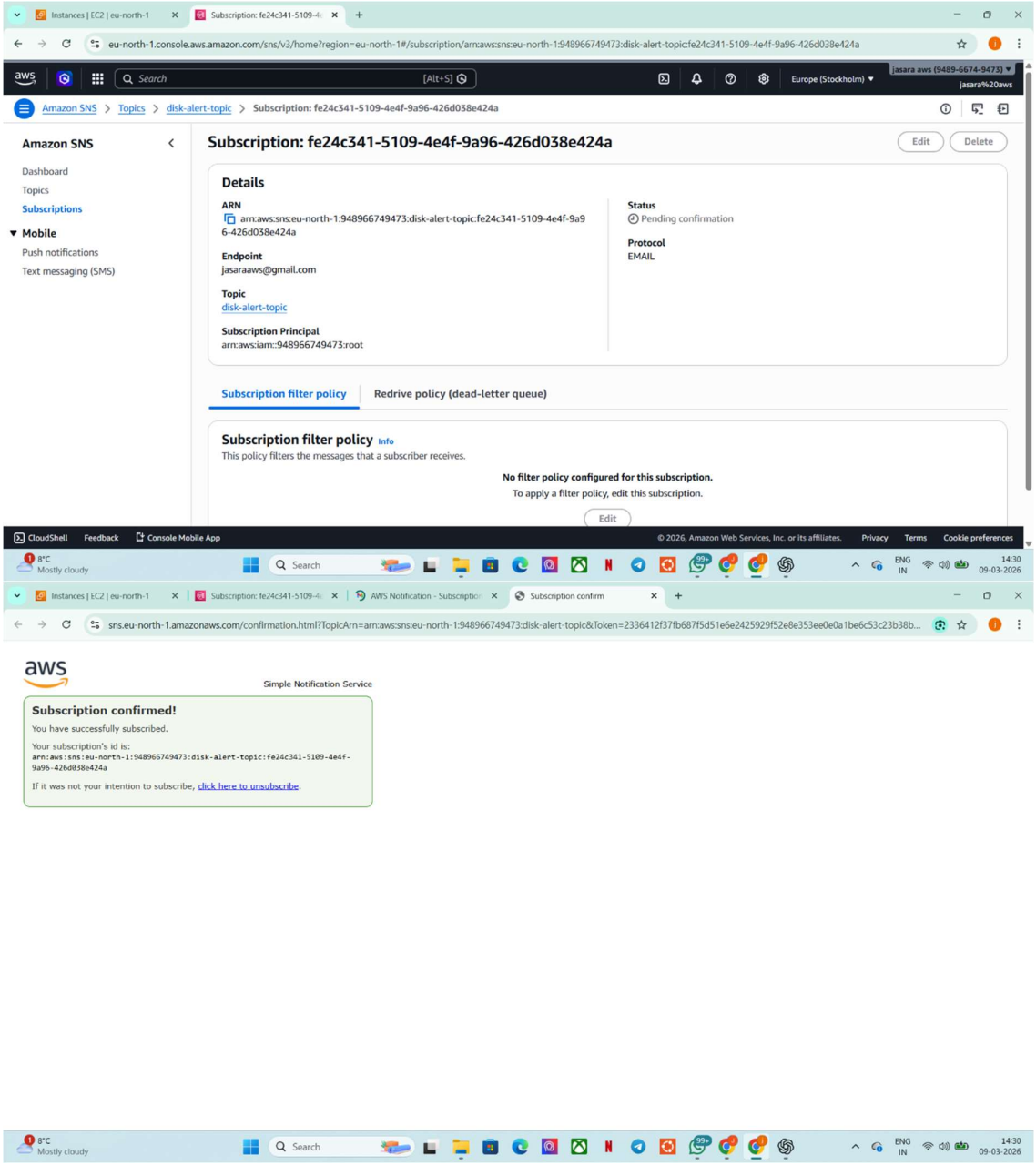
Description

1. create a script on your instance which writes the disk usage to a file and send an email to specific mail.
2. verify the recipient mail id in SNS console
3. configure simple notification service[SNS] For delivering the mail notifications.
4. verify the mail sent

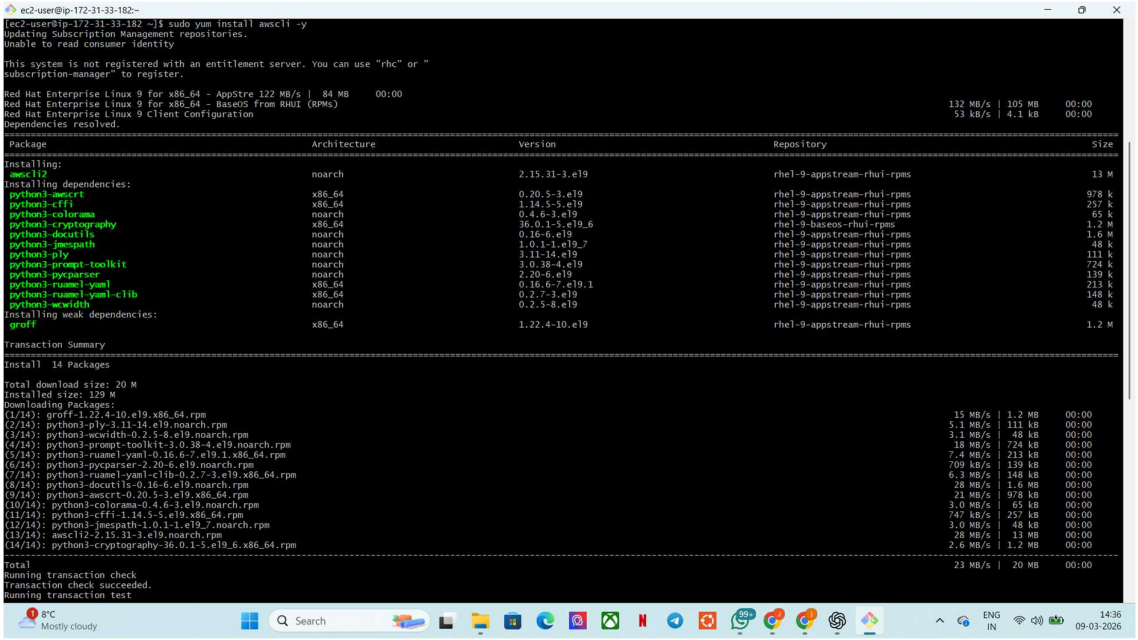
Step 1: Created an SNS Topic



Step 2: Created Subscription and Verified Recipient Email in SNS



Step 3: Installed AWS CLI on the Instance



```
ec2-user@ip-172-31-33-182:~
[ec2-user@ip-172-31-33-182 ~]$ sudo yum install awscli -y
Updating Subscription Management repositories.
Unable to read consumer identity

This system is not registered with an entitlement server. You can use "rhc" or "
subscription-manager" to register.

Red Hat Enterprise Linux 9 for x86_64 - AppStre 122 MB/s |  84 MB     00:00
Red Hat Enterprise Linux 9 for x86_64 - BaseOS from RHUI (RPMs)                  132 MB/s | 105 MB     00:00
Red Hat Enterprise Linux 9 Client Configuration                                   53 kB/s | 4.1 kB     00:00
Dependencies resolved.
================================================================================================================================================
 Package                                  Architecture                  Version                                  Repository                      Size
================================================================================================================================================
Installing:
 awscli2                                  noarch                        2.15.31-3.el9                            rhel-9-appstream-rhui-rpms      13 M
Installing dependencies:
 python3-awscrt                           x86_64                        0.20.5-3.el9                             rhel-9-appstream-rhui-rpms     978 k
 python3-cffi                             x86_64                        1.14.5-5.el9                             rhel-9-appstream-rhui-rpms     257 k
 python3-colorama                         noarch                        0.4.6-3.el9                              rhel-9-appstream-rhui-rpms      65 k
 python3-cryptography                     x86_64                        36.0.1-5.el9_6                           rhel-9-baseos-rhui-rpms        1.2 M
 python3-docutils                         noarch                        0.16-6.el9                               rhel-9-appstream-rhui-rpms     1.6 M
 python3-jmespath                         noarch                        1.0.1-1.el9_7                            rhel-9-appstream-rhui-rpms      48 k
 python3-ply                              noarch                        3.11-14.el9                              rhel-9-appstream-rhui-rpms     111 k
 python3-prompt-toolkit                   noarch                        3.0.38-4.el9                             rhel-9-appstream-rhui-rpms     724 k
 python3-pycparser                        noarch                        2.20-6.el9                               rhel-9-appstream-rhui-rpms     139 k
 python3-ruamel-yaml                      x86_64                        0.16.6-7.el9.1                           rhel-9-appstream-rhui-rpms     213 k
 python3-ruamel-yaml-clib                 x86_64                        0.2.7-3.el9                              rhel-9-appstream-rhui-rpms     148 k
 python3-wcwidth                          noarch                        0.2.5-8.el9                              rhel-9-appstream-rhui-rpms      48 k
Installing weak dependencies:
 groff                                    x86_64                        1.22.4-10.el9                            rhel-9-appstream-rhui-rpms     1.2 M

Transaction Summary
================================================================================================================================================
Install  14 Packages

Total download size: 20 M
Installed size: 129 M
Downloading Packages:
(1/14): groff-1.22.4-10.el9.x86_64.rpm                                            15 MB/s | 1.2 MB     00:00
(2/14): python3-ply-3.11-14.el9.noarch.rpm                                       5.1 MB/s | 111 kB     00:00
(3/14): python3-wcwidth-0.2.5-8.el9.noarch.rpm                                   3.1 MB/s |  48 kB     00:00
(4/14): python3-prompt-toolkit-3.0.38-4.el9.noarch.rpm                            18 MB/s | 724 kB     00:00
(5/14): python3-ruamel-yaml-0.16.6-7.el9.1.x86_64.rpm                            7.4 MB/s | 213 kB     00:00
(6/14): python3-pycparser-2.20-6.el9.noarch.rpm                                  709 kB/s | 139 kB     00:00
(7/14): python3-ruamel-yaml-clib-0.2.7-3.el9.x86_64.rpm                          6.3 MB/s | 148 kB     00:00
(8/14): python3-docutils-0.16-6.el9.noarch.rpm                                    28 MB/s | 1.6 MB     00:00
(9/14): python3-awscrt-0.20.5-3.el9.x86_64.rpm                                    21 MB/s | 978 kB     00:00
(10/14): python3-colorama-0.4.6-3.el9.noarch.rpm                                 3.0 MB/s |  65 kB     00:00
(11/14): python3-cffi-1.14.5-5.el9.x86_64.rpm                                    747 kB/s | 257 kB     00:00
(12/14): python3-jmespath-1.0.1-1.el9_7.noarch.rpm                               3.0 MB/s |  48 kB     00:00
(13/14): awscli2-2.15.31-3.el9.noarch.rpm                                         28 MB/s |  13 MB     00:00
(14/14): python3-cryptography-36.0.1-5.el9_6.x86_64.rpm                          2.6 MB/s | 1.2 MB     00:00
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Total                                                                             23 MB/s |  20 MB     00:00
Running transaction check
Transaction check succeeded.
Running transaction test
```
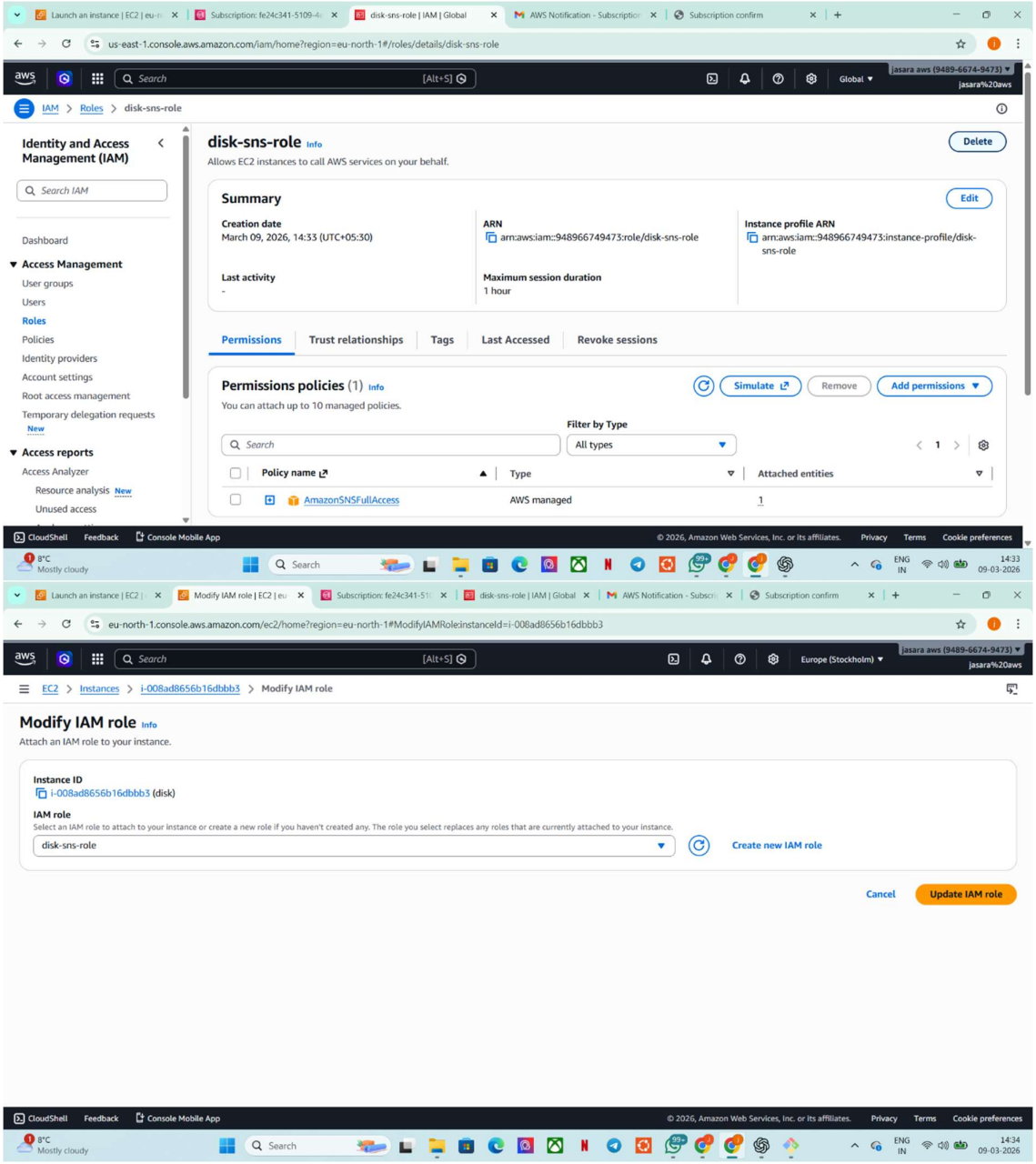
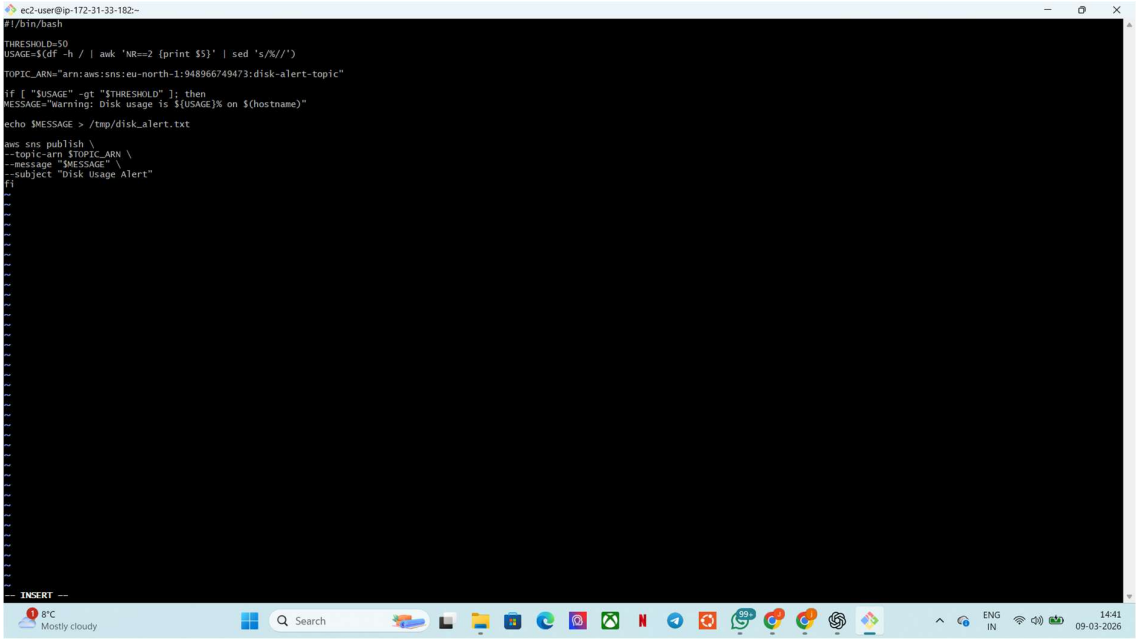

Step 4 : Created an IAM Role and Attached IAM Role to Instance

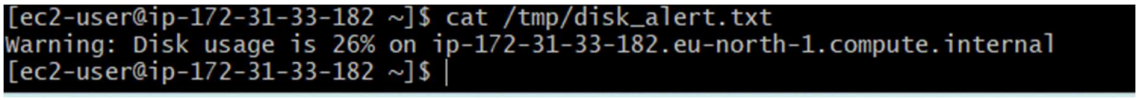
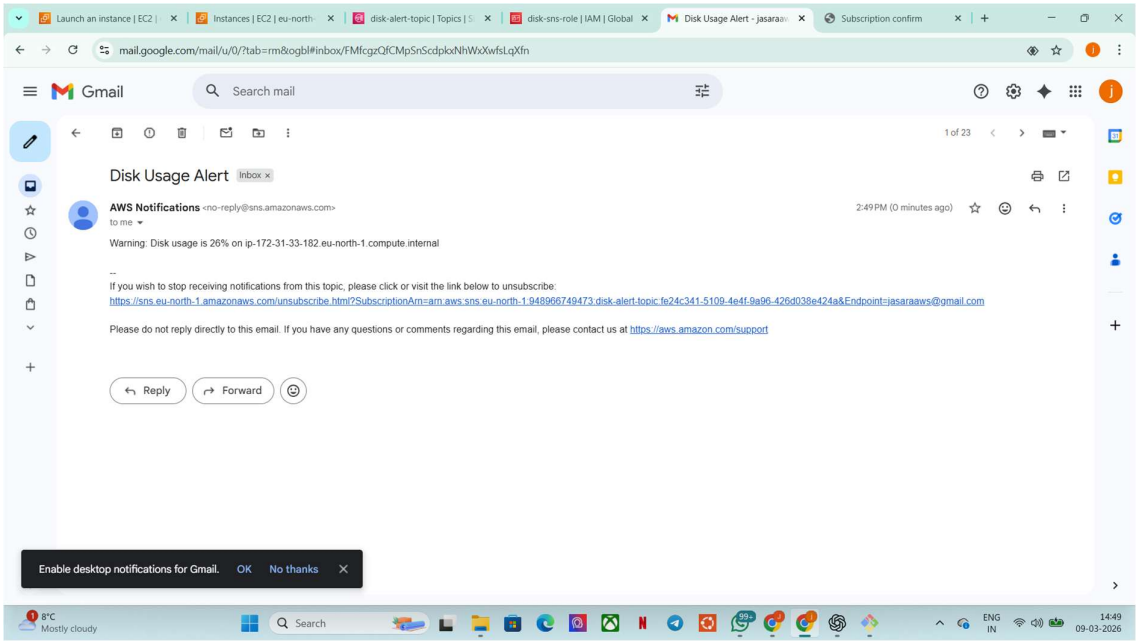


Step 5: Created Disk Usage Monitoring Script

```
[ec2-user@ip-172-31-33-182 ~]$ vi disk.sh
[ec2-user@ip-172-31-33-182 ~]$ chmod +x disk.sh
[ec2-user@ip-172-31-33-182 ~]$ ./disk.sh
[ec2-user@ip-172-31-33-182 ~]$
```



Step 6: Verified Disk Usage File



Step 7: Verified Email Notification